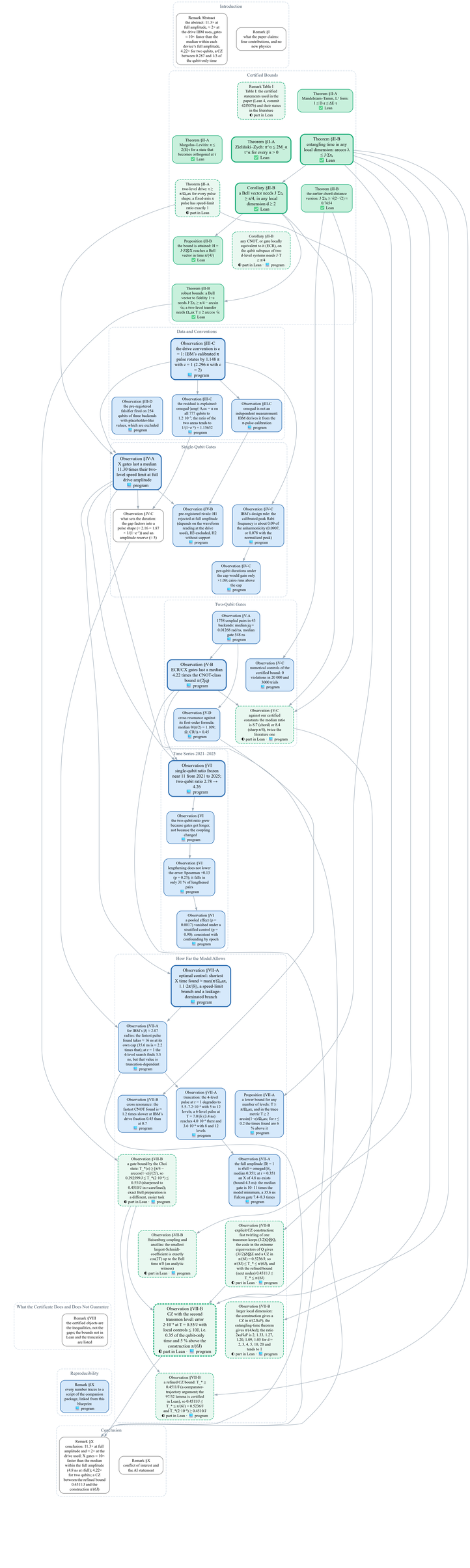
Map of *Certified Speed-Limit Diagnostics for Deployed Superconducting Quantum Processors*

Quantum speed limits · superconducting-qubit gate calibration

47 results · ✅ 8 entirely in Lean · ◐ 10 partly in Lean · 🟦 31 checked by a program · written proof only (a result can carry several tags)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



What the paper contributes

Two kinds of results, kept apart on this map. **Certified inequalities** (green, linked to Lean): the Margolus–Levitin bound, an L^1 form of Mandelstam–Tamm, the Zieliński–Zych family for every exponent $\alpha>0$, time-dependent two-level bounds, and an entangling-time bound $J\,T\ge\arccos\lambda$ in arbitrary local dimension (a Bell state needs $J\,T\ge\pi/4$, attained; robust versions with state infidelity ε). **Diagnostics on deployed hardware** (grey, with the program that computes each number): how far the calibrated gates of IBM superconducting processors are from those bounds, from public, version-pinned calibration snapshots.

- Single-qubit X gates last a median 11.30 times their two-level speed limit $\pi/\Omega_{\max}$ (1646 qubits, 41 backends), a ratio frozen near 11 from 2021 to 2025.
- Two-qubit ECR/CX gates last a median 4.22 times the known CNOT-class bound $\pi/(2j_q)$; the ratio grew from 2.78 (2021) to 4.26 (2025) because gates got longer, not because couplings got weaker.
- Calibrated drives sit at fixed fractions of an energy splitting: $\Omega_{\rm peak}\approx0.09|\delta|$ (one qubit), $\Omega_{\rm CR}\approx0.45\Delta$ (two).
- The drive-strength convention is fixed by IBM's calibrated π pulses; the $\approx15\%$ residual is explained by the pulse normalization: $\texttt{omegad}\cdot|\mathrm{amp}|\cdot A_{\rm rec}=\pi$ on all 777 qubits to $1.2\cdot10^{-7}$, and the residual tends to $1/(1-e^{-2})=1.15652$.
- At the drive IBM actually uses the X gate is within about 2 of the square-pulse limit (shape factor 1.87, or 2.16 with Qiskit's peak normalization); the 11.3 is against full drive amplitude, which IBM does not use.
- Optimal control on IBM's own transmon model finds feasible X gates of ≈16 ns at IBM's amplitude cap and 4.8 ns at its full amplitude (minimum at least 4.3 ns): 7.4–8.3 times faster than the archived 35.6 ns within the same amplitude budget; ≈3.4 ns needs three times that amplitude (numerical witnesses, not physical floors; the 6-level pulse is converged to 12 levels). The fit $\max(\pi/\Omega_{\max},\,\sim1.1\cdot2\pi/|\delta|)$ is empirical, not a theorem; $T\ge\pi/\Omega_{\max}$ (and $2\arcsin(1-e)/\Omega_{\max}$ in the trace metric) holds for any number of levels.
- With the second transmon level the minimal CZ time lies between 0.287 and $1/3$ of the qubit-only time: $0.4511/J\le T_*\le\pi/(6J)$ (refined bound, explicit construction; the certified norm bound is $\pi/(8J)$); optimal control reaches error $2\cdot10^{-8}$ at $0.55/J$ (0.35), and for the same task $0.4510/J\le T_*(2\cdot10^{-8})\le0.55/J$. In larger local dimension the construction and the entangling-time theorem approach each other (ratio $\to1$, certified in Lean).

The paper claims no new physics: the speed limits are classical, the two-qubit constants are known for qubits, and the transmon model is IBM's. The entangling-time bound is, for qubits, a known corollary of Dür *et al.* and Vidal–Hammerer–Cirac, new only as a machine-checked, any-dimension assembly. This blueprint is built from a **draft**: the paper's own open items are shown where they matter.

🚀 **Start here:** **Observation §IV-A**: X gates last a median 11.30 times their two-level speed limit at full drive amplitude.

Main results:

- **Theorem §II-A**: Zieliński–Zych: $\pi^\alpha \le 2M_\alpha \tau^\alpha$ for every $\alpha > 0$

- **Theorem §II-B**: entangling time in any local dimension: arccos $\lambda \le J \cdot \Sigma \tau_k$
- **Corollary §II-B**: a Bell vector needs $J \cdot \Sigma \tau_k \ge \pi/4$, in any local dimension $d \ge 2$
- **Observation §III-C**: the drive convention is $c = 1$: IBM's calibrated π pulse rotates by 1.148 π with $c = 1$ (2.296 π with $c = 2$)
- **Observation §IV-A**: X gates last a median 11.30 times their two-level speed limit at full drive amplitude
- **Observation §V-B**: ECR/CX gates last a median 4.22 times the CNOT-class bound $\pi/(2jq)$
- **Observation §VI**: single-qubit ratio frozen near 11 from 2021 to 2025; two-qubit ratio 2.78 → 4.26
- **Observation §VII-A**: optimal control: shortest X time found $\approx \max(\pi/\Omega_{max}, 1.1 \cdot 2\pi/|\delta|)$, a speed-limit branch and a leakage-dominated branch
- **Observation §VII-B**: CZ with the second transmon level: error $2 \cdot 10^{-8}$ at $T = 0.55/J$ with local controls $\le 10J$, i.e. 0.35 of the qubit-only time and 5 % above the construction $\pi/(6J)$

🧭 Fields: where the paper lives and what it connects

Tools taken from other branches

- **Quantum speed limits** → Margolus–Levitin, Mandelstam–Tamm and the Zieliński–Zych family, applied to the drive amplitude of a transmon and to the coupling of a pair — Theorem §II-A, Theorem §II-A, Theorem §II-A, Theorem §II-A
- **Entangling capabilities of Hamiltonians** → minimal interaction time under unbounded local control; Schmidt coefficients and the Fubini–Study angle (Dür et al., Vidal–Hammerer–Cirac, Rudnicki) — Theorem §II-B, Corollary §II-B, Proposition §II-B, Corollary §II-B
- **Formal verification** → Lean 4 + Mathlib: the inequalities of Table I — Theorem §II-A, Theorem §II-A, Theorem §II-A, Theorem §II-B, Corollary §II-B, Proposition §II-B, Theorem §II-B
- **Optimal control** → piecewise-constant GRAPE control on a 4-level transmon with a bound on the drive — §VII-A, §VII-A, §VII-A, §VII-A, §VII-B

What the results serve

- **Gate design on superconducting processors** ← how far calibrated gates are from the physical limit, and which fractions of the energy splittings the calibrations use — §IV-A, §V-B, §IV-C, §V-D
- **Reproducible hardware benchmarking** ← version-pinned public calibration snapshots as a citable source, and a stratified control against confounding by epoch — §VI, §VI